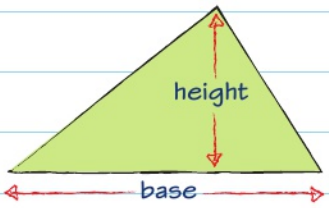
Formulae

A **formula** is a mathematical rule.
Formulae is the plural of formula.
The formula for the area of this triangle is:

Area = $\frac{1}{2}$ × Base × Vertical height

You can write this formula using algebra as:

$A = \frac{1}{2}bh$



A formula lets you calculate one quantity when you know the others.
You need to **substitute** the values you know into the formula.

For more on substitution have a look at page 25.

Worked example

You can use this formula to work out the cooking time in minutes for a turkey.

Cooking time = Weight in kg × 30 + 45

Work out the cooking time for a turkey weighing 7 kg. **(3 marks)**

Cooking time = 7 × 30 + 45
= 210 + 45
= 255

The cooking time is 255 minutes, or 4 hours and 15 minutes.

Substitute the weight into your formula before you do any calculations.

Remember to use **BIDMAS** for the correct order of operations. You **M**ultiply before you **A**dd.

For a reminder about substituting and BIDMAS have a look at page 25.

When you are giving an answer in minutes and it is larger than 60 minutes, you can give it in minutes or in hours and minutes.

4 × 60 = 240 and 255 − 240 = 15

So 255 minutes = 4 hours and 15 minutes.

Worked example

This formula is used in physics to calculate distance.

$D = ut - 5t^2$
$u = 20$
$t = 3$

Work out the value of D. **(3 marks)**

$D = 20 \times 3 - 5 \times 3^2$
$= 20 \times 3 - 5 \times 9$
$= 60 - 45$
$= 15$

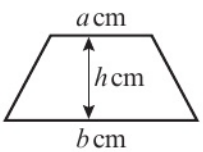
Substitute the values for u and t into the formula. Using **BIDMAS**:

1. Do the **I**ndex (power) first.
   $3^2 = 9$
2. Do the **M**ultiplications next.
   $20 \times 3 = 60$
   $5 \times 9 = 45$
3. Do the **S**ubtraction last.
   $60 - 45 = 15$

Don't try to do more than one operation on each line of working.

Now try this

1 Here is a trapezium.
The area of the trapezium is given by the formula Area = $\frac{1}{2}(a + b)h$
Work out the area when
$h = 4.5$, $a = 6.5$ and $b = 9.5$ **(3 marks)**



2 Here is a formula: $D = b^2 - 4ac$
Work out the value of D when
$a = 2$, $b = 5$ and $c = -3$ **(3 marks)**

Substitute all the values before you do any calculations.

Writing formulae

You can write a rule given in words as a word formula or as a formula using algebra.

This label gives instructions for working out the cooking time of a chicken.

FREE-RANGE CHICKEN		
WEIGHT (KG) 1.8	PRICE PER KG £3.95	COOKING INSTRUCTIONS Cook at 170°C for 25 minutes per kg plus half an hour

You can write the cooking instructions as a word formula.

Cooking time in minutes = 25 × Weight in kg + 30

You need to give units when you are describing the quantities in a formula.

If the cooking time is in hours then this formula would give you a very crispy chicken!

You can also write this formula using algebra.

$T = 25w + 30$, where T is the cooking time in minutes and w is the weight in kg.

When you write a formula using algebra you need to explain what each letter means.

Worked example

Chloe buys a pens costing 25 pence each and b pencils costing 15 pence each. Write a formula for the total cost T pence. **(2 marks)**

$T = 25a + 15b$

$25a$ means $25 \times a$. You should make sure your formula is simplified as much as possible. Don't write any units (like pence) in your formula.

Worked example

The cost of hiring a car can be worked out using this formula.

Cost = £80 + 50p per mile

Write a formula for the cost £C of hiring a car which is driven for m miles. **(2 marks)**

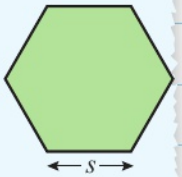
$C = 80 + 0.5m$

You need to make sure all your values are in the same units. C is the cost in pounds, so convert 50p into pounds. 50p = £0.50, so you need to multiply the number of miles by 0.5

Worked example

The diagram shows a regular hexagon.

Write a formula for the perimeter of the hexagon P in terms of s. **(1 mark)**



$P = 6s$

The perimeter is the distance all the way around a shape.
All the sides of a regular shape are the same length. There are 6 sides so:
Perimeter = 6 × length of one side
$P = 6s$

You can find out more about perimeter on page 79.

Now try this

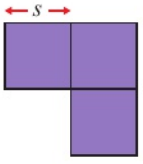
1 Rulers cost 50 pence each and pens cost 35 pence each.

(a) Write an expression for the total cost in pence of x rulers and y pens. **(1 mark)**

(b) Complete this formula for the total cost, T pence.

$T =$ _____ **(1 mark)**

2 This shape is made from 3 identical squares.



(a) Write a formula for the area, A, of the shape. **(1 mark)**

(b) Work out the area of the shape when $s = 5$ cm. **(1 mark)**

Worked solution video

Expanding brackets

Expanding brackets is sometimes called **multiplying out** brackets.

Golden rule

You have to multiply the expression outside the brackets by everything inside the brackets.

$4n \times n = 4n^2$

$$4n(n + 2) = 4n^2 + 8n$$

$4n \times 2 = 8n$

For a reminder about multiplying expressions look back at page 23.

You need to be extra careful if there are negative signs outside the brackets.

$-2 \times x = -2x$

$$-2(x - y) = -2x + 2y$$

$-2 \times -y = 2y$

Multiplying negative terms is like multiplying negative numbers.

When both terms are + or both terms are − the answer is **positive**.

$-2a \times -a = +2a^2$

When one term is + and one term is − the answer is **negative**.

$-10p \times 5q = -50pq$

Sometimes you have to **expand and simplify**. This means 'multiply out the brackets and then collect like terms'.

$6 \times m = 6m$ $\quad 4 \times 3 = 12$

$$6(m + 2) + 4(3 - m) = 6m + 12 + 12 - 4m$$

$6 \times 2 = 12$ $\quad 4 \times -m = -4m$

$$= 2m + 24$$

$6m - 4m = 2m$ $\quad 12 + 12 = 24$

Remember that any negative signs belong to the term on their right.

Look back at page 22 for a reminder about collecting like terms.

Worked example

(a) Expand $5(2y - 3)$ **(2 marks)**

$5(2y - 3) = 10y - 15$

(b) Expand and simplify $2(3x + 4) - 3(4x - 5)$ **(3 marks)**

$2(3x + 4) - 3(4x - 5)$
$= 6x + 8 - 12x + 15$
$= -6x + 23$

For part (b), be careful with the second bracket.
$-3 \times 4x = -12x$
$-3 \times -5 = +15$
The question says 'expand and simplify' so remember to collect any like terms after you have multiplied out the brackets.

Now try this

1 Expand (a) $3(y - 6)$ **(2 marks)** (b) $m(m + 7)$ **(2 marks)**

2 Expand and simplify
(a) $5(a - 2b) + 4(2a + b)$ **(3 marks)**
(b) $2(4w + 3) - 3(3w - 5)$ **(3 marks)**
(c) $2m(m + 9) - m(m - 4)$ **(3 marks)**

Be careful when there is a minus sign in front of the second bracket.
$-3 \times 3w = -9w$ and
$-3 \times -5 = 15$